SEAGAS - Team Update: What Is Done, What Is Left, What To Do Next

From Carl (Soh Way Miin), Backend Developer, to Group 10 | 30 October 2026
COS40005 Computing Technology Project A | POC004

In short: students can now **upload a CV** (SEAGAS picks up the skills in it) and **export a CV**; they can list **work experience** and see **Career Paths** (FR-15). The **advisor portal** is complete: charts, filters, CSV export, learning progress, **advisor notes** with reviews of AI results, CV download and a printable report. Admins have **System Settings** (FR-16), and SEAGAS can **send real emails** once someone adds mail-server settings to `backend/.env` (section 3). After you pull, run **start_all.bat** once.

Note: these changes go straight to dev, without a pull request. The new pages have passed the automated tests (116) but have **not been checked in a browser yet**; see section 2.2.

1. What Is Done (30 October)

1.1 Students

Feature	What it does
CV upload	My Profile → Skills: upload a PDF or Word (.docx) CV. The NLP module (hybrid) finds the skills it mentions **that are in the skills library**; other skills are not picked up. Skills outside the student's course list are shown but cannot be added. Nothing is added until the student ticks the skills and chooses a level. One CV per student (a new upload replaces it); download or remove it any time. Scanned (image-only) PDFs are refused with an explanation.
Export CV	My Profile → **Export CV**: a one-page CV with details, skills by category (level shown as ●●○), work experience, projects, certifications and completed learning. **Save as PDF** uses the browser's print dialog.
Work experience	My Profile → **Experience**: internships, part-time / full-time jobs, freelance work and volunteering, with dates ("I still work here"), location and a description. Every type except volunteering now counts towards the **Industry Experience** score of an assessment.
Career Paths (FR-15)	New menu item. Ranks the job roles of the student's course by skill match (weights from Aaron's engine: required 1.0, preferred 0.6, bonus 0.3), shows what the student has and what is missing, and suggests a free resource for up to three missing required skills. Buttons: **Assess me against this role** (opens Job Analysis with the role chosen) and **Make this my target role**.

1.2 Advisors

Area	What it does
Overview	Readiness distribution chart, student status (On Track / Developing / **Needs Support** / not assessed, click to filter the student list), top skill gaps, target roles with average readiness, learning activity totals.
Skill Gaps	Filter by the job role the students were assessed against; export CSV.
Students	Search and filters (status, programme, target role, profile complete), sortable columns, last assessed, activities completed, number of notes; export the filtered list as CSV.
Learning Progress (new tab)	Completed / in-progress / saved activities, the resources completed most often, and each student's activity; export CSV.

Student page	**Advisor notes**: general notes and **reviews of the latest AI result** (agree / disagree, with a reason). Seen by staff of the course only, **never by the student**; only the author edits or deletes; all audited. Also: **Download CV** (audited), work experience, GPA / CGPA, **Print report**.
Human oversight	A "Decision support, not a decision" notice on the advisor pages; "At Risk" is now "Needs Support" everywhere. Useful evidence for the ethics / human-oversight part of the report.

1.3 Admins

Area	What it does
System Settings (FR-16)	New menu item. Semester start dates (GPA / CGPA reminder), readiness bands for the advisor dashboard (On Track / Needs Support limits, with a preview), largest upload (1-5 MB) and evidence per item (1-10). Changes apply at once and are audited; values are checked (e.g. the Needs Support limit must be below On Track).
Email	System Settings shows whether a mail server is set up (never the password) and can **send a test email**. Student Intake → Email outbox has **Resend** for each email, with the number of attempts and when it was sent. An old temporary password is not resent once the person has chosen their own password.
Audit log	New actions: advisor notes, advisor CV downloads, settings changes, email resends.

1.4 Also since the last update (29 October)

Change	Details
Programmes	13 in total. Computer Science: Software Engineering, Data Science, Artificial Intelligence, Cybersecurity, Cloud Computing. Information and Communications Technology: Web and Mobile Development, Data Management, Network Security, Cloud and Networking. Business: Business Analytics, Marketing, Finance, Human Resource Management. Old names (Software Development, Software Systems, Information Systems, Digital Marketing, Accounting) are merged by the migration, also after every synthetic data load.
Pages	"ICT" is written in full; wide tables scroll inside their card instead of cutting off the page; IDs, emails and table headings stay on one line; the admin Skills & Roles library is filtered by course.
Gold set	The gold workbook (annotator A) was filled in; Yong Hin collected it, measured agreement with annotator B and froze `gold_labels.csv`.

1.5 Database and tests

New tables: `student_cvs`, `advisor_notes`, `student_experiences`, `system_settings`; `email_outbox` gained `attempts`, `last_attempt_at` and `sent_at`. All are in `schema.sql` and `migrate_existing_db.sql`. New Python package: `pypdf` (reads PDF CVs), installed by the start script. Tests: 116 pass (`python -m pytest tests -q`); the email tests use a fake mail server, so nothing is really sent.

2. What To Do Next

2.1 After you pull (every member)

1. `git checkout dev`, `git pull`, then double-click **start_all.bat**. It installs `pypdf`, creates the new tables and renames the old programmes.
2. Test student: `student0001@synthetic.example.com` / `900000001` / `Seagas@2026`. Try Career Paths, the Experience tab, CV upload (any made-up CV) and Export CV.
3. As an advisor, open the four dashboard tabs and a student's page; add a note and a review of the AI result.
4. As an admin, open System Settings. To send real emails, see section 3.
5. Use only made-up names, emails, CVs and phone numbers (NFR-05).

2.2 Please check in the browser (Ren Hang, Shanice)

- Student: Experience tab, Career Paths, CV upload (PDF and Word), Export CV → Save as PDF.
- Advisor: Overview charts, filters and sorting, CSV exports open in Excel, notes add / edit / delete, Print report.
- Admin: System Settings (save, undo, invalid values), Email outbox Resend.
- Report anything that looks wrong or confusing to Carl.

3. Sending Real Emails (one-time setup)

Why: new students and advisors receive their login details by email. Without a mail server the emails are only stored in Student Intake → Email outbox, and the admin has to pass the details on by hand. For the demo it is much better if they really arrive.

Who: whoever runs the backend for the demo, with a group mailbox (e.g. a Gmail account made for the project). The settings live in `backend/.env` on that computer only.

1. Prepare a mailbox, for example the group's Gmail account.
2. Turn on **2-Step Verification**, then create an **App Password** (Google Account → Security → App passwords). The normal account password does not work.
3. Add these lines to `backend/.env` (this file is **not** in git — never commit it), then restart the backend:

```
SMTP_HOST=smtp.gmail.com
SMTP_PORT=587
SMTP_SECURITY=starttls
SMTP_USER=your.mailbox@gmail.com
SMTP_PASSWORD=the 16-letter App Password
SMTP_FROM=your.mailbox@gmail.com
```

4. Sign in as an admin, open **System Settings** and press **Send test email** to check that it arrives (look in the spam folder too).
5. Go to **Student Intake** → **Email outbox** and **Resend** the emails that were stored before.

Other providers give their own host, port and security; SSL uses `SMTP_PORT=465` with `SMTP_SECURITY=ssl`. Do not share the App Password in chat or in the report; anyone with it can send mail as the group account.

4. What Is Still Left

Owner	Task	When
One member	Set up the group mailbox and the mail-server settings for the demo (section 3).	Before demo
Ren Hang, Shanice	Browser check of the new pages (section 2.2) and small UI fixes.	Sprint 3
Yong Hin	Check the Business role templates against Business JDs; optional: the synthetic data generator can write the new programme names directly (the migration renames them anyway).	Sprint 3-4
Aaron	Check the Business learning resources and cost tiers; Career Paths uses the weights of your engine function.	Sprint 3
Whole team	SUS usability test; final report (design, implementation, testing, ethics / human oversight) and demo.	Sprint 4